

REGRESSION ANALYSIS (LECTURE 2)



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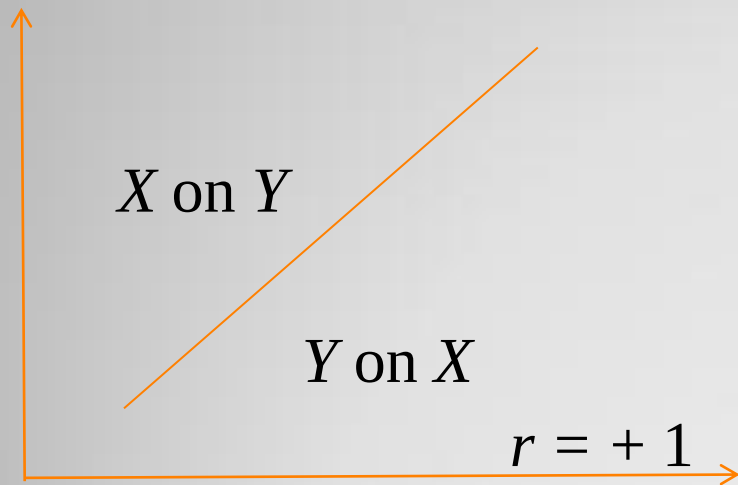
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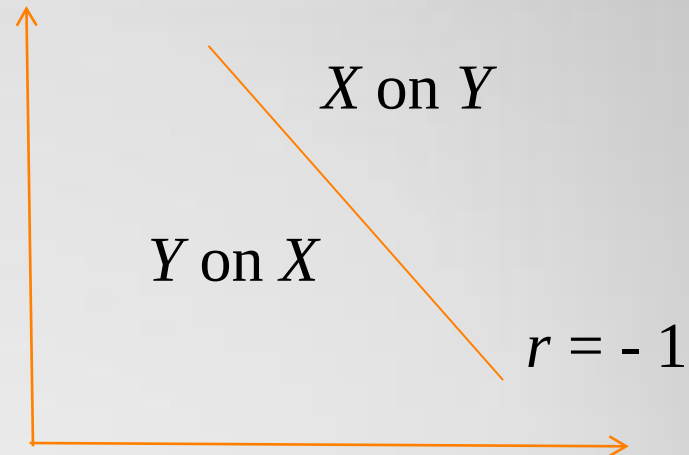
Simple Linear Regression Lines:

The equation of Regression line of Y on X : $y = a + bx$. Here y is the dependent variable (effect) and x is the independent variable (cause).

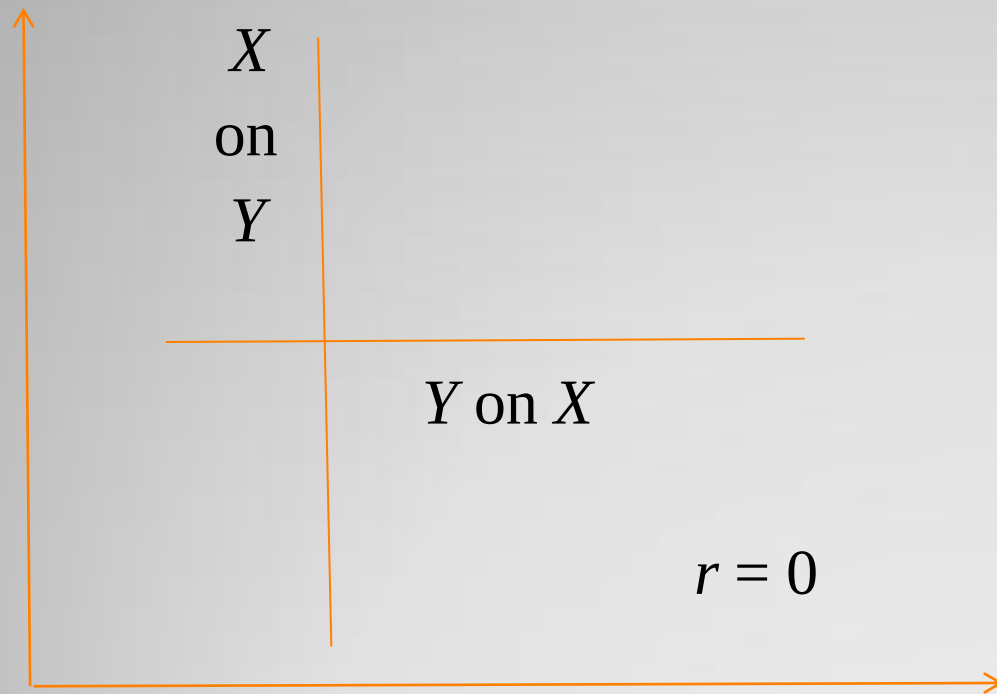
The equation of Regression line of X on Y : $x = c + dy$. Here x is the dependent variable (effect) and y is the independent variable (cause).



Perfect Positive Correlation



Perfect Negative Correlation



No Correlation

Equations of Simple Linear Regression Lines:

The equation of Regression line of Y on X:

$$y - \bar{y} = b_{yx} (x - \bar{x})$$

where b_{yx} is the regression coefficient of Y on X given by

$$b_{yx} = r \cdot \frac{s_y}{s_x}$$

The equation of Regression line of X on Y:

$$x - \bar{x} = b_{xy} (y - \bar{y})$$

where b_{xy} is the regression coefficient of X on Y given by

$$b_{xy} = r \cdot \frac{s_x}{s_y}$$

Here r is the correlation coefficient between X and Y , $\bar{x}$ is the mean of X , $\bar{y}$ is the mean of Y , s_x is the standard deviation of X and s_y is the standard deviation of Y . Moreover these two regression lines intersect each other at the point $(\bar{x}, \bar{y})$.

If $r = 0$ then the equation of regression line of Y on X : $y - \bar{y} = 0$

If $r = 0$ then the equation of regression line of X on Y : $x - \bar{x} = 0$

So for $r = 0$ i.e. when X and Y are linearly independent the corresponding regression lines are perpendicular to each other.

If $r = 1$ or -1 then in each case these two regression lines coincide. For $r = 1$ they take similar form as:

$$\frac{y - \bar{y}}{s_y} = \frac{x - \bar{x}}{s_x}$$

For $r = -1$ they take the form as

$$\frac{y - \bar{y}}{s_y} = - \frac{x - \bar{x}}{s_x}$$

Also the product between two regression coefficients is $b_{yx} \cdot b_{xy} = r^2$

The quantities b_{yx} , b_{xy} and r possess the same sign.

In general the angle between two regression lines is given by

$$\begin{aligned} & \tan^{-1} \left| \frac{\frac{1}{b_{xy}} - b_{yx}}{1 + \frac{b_{yx}}{b_{xy}}} \right| \\ &= \tan^{-1} \left| \frac{1 - b_{yx} \cdot b_{xy}}{b_{xy} + b_{yx}} \right| \\ &= \tan^{-1} \left| \frac{1 - r^2}{r \cdot \left(\frac{s_x}{s_y} + \frac{s_y}{s_x} \right)} \right| \\ &= \tan^{-1} \left| \frac{(1 - r^2) \cdot s_x \cdot s_y}{r \cdot (s_x^2 + s_y^2)} \right| \end{aligned}$$

Problem: For two variables X and Y the two regression lines are given by $x - 3y + 1 = 0$ and $5x - y = 23$. Find the mean of X and Y . Also find the correlation coefficient between X and Y .

Answer: For two variables X and Y the two regression lines are given by $x - 3y + 1 = 0$ and $5x - y = 23$. Solving we get $x = 5$ and $y = 2$. As two regression lines intersect each other at the point with abscissa as mean of X and ordinate as mean of Y so here mean of X is 5 and mean of Y is 2. The first equation can be written as $y = \frac{1}{3}(x + 1)$ or, $y - 2 = \frac{1}{3}(x - 5)$. So $b_{yx} = \frac{1}{3}$. Again, the second equation can be written as $x = \frac{1}{5}(y + 23)$ or, $x - 5 = \frac{1}{5}(y - 2)$. So $b_{xy} = \frac{1}{5}$. Here $b_{yx} \cdot b_{xy} = r^2$. So $r^2 = \frac{1}{3} \cdot \frac{1}{5} = \frac{1}{15}$. Hence, $r = \frac{1}{\sqrt{15}}$

[Here as b_{yx} and b_{xy} are both positive so r is also positive]

Note: We must choose the regression lines Y on X and X on Y in such a way so that

$$b_{yx} \cdot b_{xy} = r^2 \leq 1$$